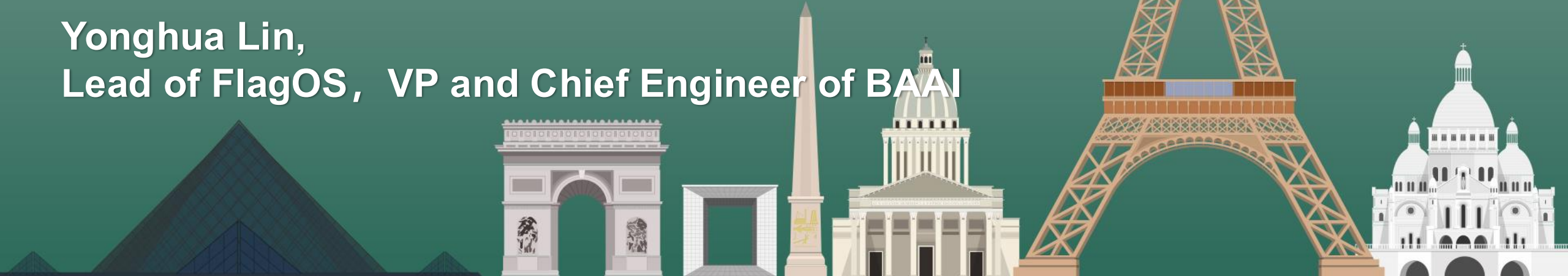


From ChatGPT to OpenClaw: How Agentic AI is Redefining AI Compute and the Open Evaluation

A DeepSeek V4 Multi-Chip Adaptation Story with FlagOS

Yonghua Lin,
Lead of FlagOS, VP and Chief Engineer of BAAI



From ChatGPT to OpenClaw: AI Competition Moves to Compute Infrastructure

ChatGPT: Large-model era

- Models enter public & industry use
- Compute becomes core R&D capital
- Competition entered to model capability

OpenClaw: Agentic era

- AI perceives, plans, calls tools, and acts
- Inference becomes production compute
- Competition shifts to compute infrastructure

Large models opened the AI era; agents will reshape the real world.

The next bottleneck is not only model capability — it is computing infrastructure.

Important open-weight models released in 2026

Model/Organization	Model information
DeepSeek-V4 / DeepSeek	DeepSeek-V4-Pro (1.6T params, 49B active), DeepSeek-V4-Flash (284B parms, 13B active)
Qwen 3.5 & 3.6 / Alibaba (Qwen)	Qwen3.5-397B-A17B, Qwen3.5-35B-A3B, Qwen3.6-27B, Qwen3.5-Omni
Gemma 4 / Google DeepMind	Gemma 4 (2B, 4B, 26B-A4B, 31B)
MiniCPM / ModelBest	MiniCPM-o 4.5 (9B)
MiniMax M2.5 & M2.7 / MiniMax	MiniMax M2.5 (Agent-focused), MiniMax M2.7 (Self-evolving)
Kimi K2.5 & K2.6 / Moonshot AI	Kimi-K2.5 (terascale MoE), Kimi-K2.6
GLM-5.1 / Z.ai	GLM-5.1
Nemotron 3 / NVIDIA	Nemotron 3 Super, Omni, VoiceChat
Granite 4.1 / IBM	Granite 4.1 (3B, 8B, 30B)
Xiaomi MiMo V2.5 / Xiaomi	MiMo-V2.5, MiMo-V2.5-Pro
Sarvam 30B / 105B / Sarvam AI	Sarvam-30B, Sarvam-105B
SenseNova U1 / SenseTime	SenseNova-U1
Mistral Medium 3.5 / Mistral AI	Mistral-Medium-3.5-128B
Hunyuan HY-3.0 / Tencent	HY-3.0-preview (295B total, 21B active)
Phi-4 / Microsoft	Phi-4-Reasoning-Vision-15B

For the reasons of
Security · Control · Privacy · Resilience · Innovation



More chips

AI systems can run on more different available chipsets



More choices

Developers can choose the right hardware for the right workload



Lower costs

Agentic AI can move from demos to real-world deployment

Open compute turns open-weight models into real deployment choices.

Let's use DeepSeek V4 as example

DeepSeek V4 shows why frontier open models need open compute.

Why DS-V4 matters for

Agentic AI

- 1M-token context
- MoE architecture
- Tool calling and interleaved thinking



What DS-V4 requires

- Hybrid CSA/HCA attention
- Heterogeneous KV cache
- FP4 / FP8 precision
- New kernels and runtime optimization



Why deployment becomes hard?

- Only chips top-end AI chips supports FP4/FP8 precision.
- Requiring >80GB memory for single device.
- Kernels remain coupled with specific hardware backends.

**DeepSeek V4 shows why open compute matters:
frontier open models need a portable path to deployment across chips.**

What should we do to migrate DeepSeek V4 onto different chips?

- Replace hardware-bound components with a **unified cross-chip software stack**
- Implement **missing operators** for DeepSeek V4 (5 new ones)
- **Convert unsupported precision** formats for non-Blackwell hardware
- Redesign tensor parallelism for **memory-constrained** devices
- **Optimize communication** for multi-node, multi-chip deployment
- **Validate** numerical correctness and benchmark accuracy
- **Tune performance** for each hardware backend
- Release ready-to-use converted models and runnable examples

FlagOS: An Open Compute Stack for the Multi-Chip Era

 What is FlagOS? <https://github.com/flagos-ai>

FlagOS 2.0 is a unified, open-source AI system software stack that enables AI applications to run seamlessly across diverse hardware platforms.

18
Chip Vendors

32
AI Chips

Various AI models

Language models

Multimodal models

World models

Embodied AI

FlagOS Ecosystem-enabled modules

Megatron-FL

TransformerEngine-FL

vLLM-plugin-FL

VeRL-plugin-FL

SGLang-plugin-FL

Pytorch-plugin-FL

Domain solution

FlagOS-Robo

FlagOS Skills

Deploy, code-gen, migration, benchmark

FlagOS core libraries

FlagGEMs operator libraries

FlagGEMs, FlagBLAS, FlagSparse, FlagFFT, FlagTensor, FlagAudio

FlagScale parallel framework

FlagTree Unified AI compiler

FlagCX Unified comm. lib

FlagOS tools

KernelGen: operator auto-gen tool

FlagRelease: Model auto-migration tool

FlagPerf: AI chip evaluation tool

Servers of different AI chipsets



Edge devices



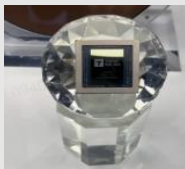
AI book



ARM devices



RISC-V AI chips

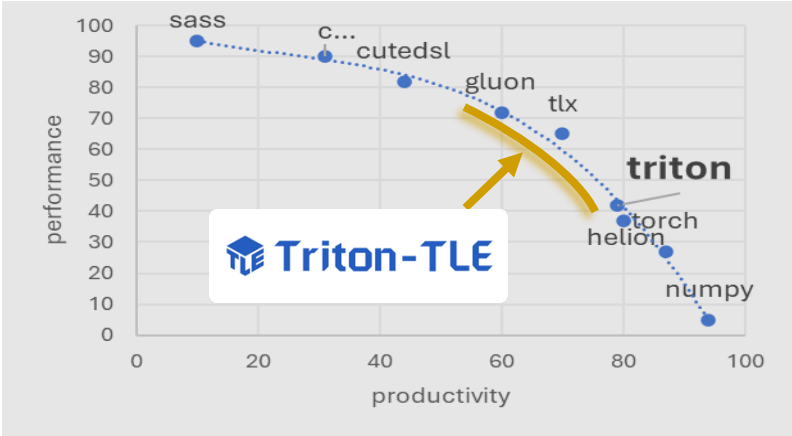


Key technologies to enable DeepSeek V4 onto various AI chips

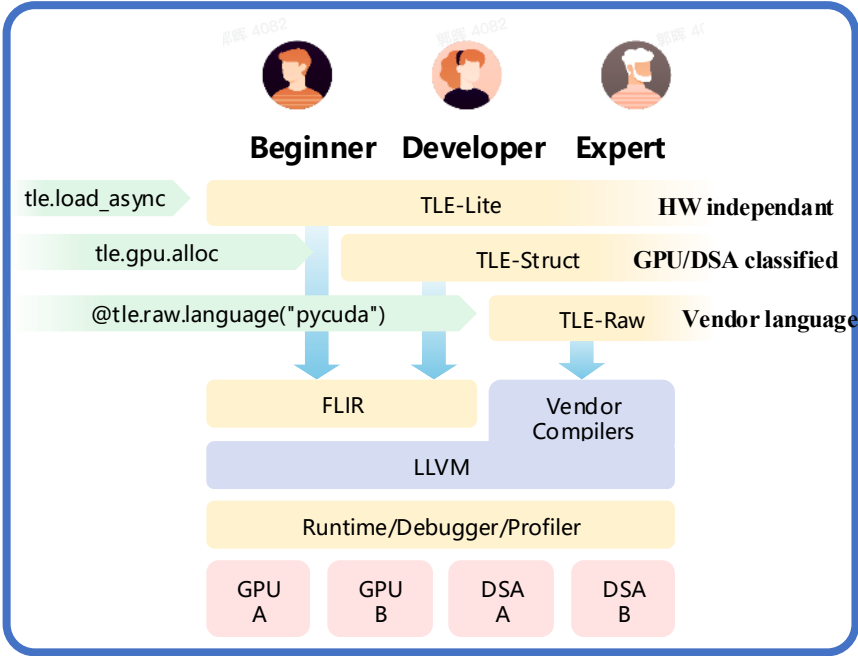
- **FlagGems:** full operator replacement
- **KernelGen:** rapid generation of new operators
- **FlagTree / Triton-TLE:** write once, compile across chips
- **vLLM-plugin-FL:** multi-chip inference integration
- **FlagCX:** o-group parallelism, communication & parallel strategy
- **FlagRelease:** release once, run across chips

FlagTree: Unified Compiler for Various AI chips and supporting Triton-TLE New Programming Language

Triton-TLE (Triton Language Extension) is compatible with standard Triton while providing three levels of language extensions. It maintains the ease-of-use advantages of Tile programming while significantly enhancing chip architecture awareness and dramatically improving performance.



Supported 12 different AI chips



Triton-TLE Performance Improvements (vs Native Triton)

Operator	Chip	Performance Gain
MoeAlignBlockSize	NVIDIA	25%~200% improvement, up to 4x CUDA
TopK	NVIDIA	1.6x improvement
SparseMLA	NVIDIA	27% improvement
SparseMLA	Huawei Ascend	82x improvement, 1.24x native Ascend C
RMSNorm+RoPE	Huawei Ascend	2.37x improvement, on par with native
GEMM	TsingMicro	1.5x improvement

FlagGems: the Largest Multi-Chip Triton Operator Library, 500+ Operators

FlagGems: 430+ AI operators

Largest single Triton operator library

- FlagGems is a PyTorch Foundation ecosystem proj.



- 18 important vLLM fused operators supported

Inference operator coverage

90%–100% coverage across 40 mainstream model inference tasks

C++ Wrapper adopted by multiple chip vendors to reduce Triton invocation overhead

6 new Triton domain libraries: 90 operators

FlagOS expands from large model-specific operators to broader scientific computing coverage.

FlagDNN Deep neural network 10	FlagBLAS Basic LinAlg 30
FlagFFT Fast Fourier 2	FlagSparse Sparse 22
FlagTensor Tensor	FlagAudio Speech

Full Operator Replacement for DeepSeek V4

DeepSeek V4 originally uses ~67 CUDA + TileLang operators

- FlagGems covers all 67 operators
- MoE scheduling, Attention, RMSNorm and TopK are implemented with Triton / Triton-TLE
- C++ Wrapper reduces Triton launch overhead
- Supports 10 classes of chip backends for DeepSeek V4

Supports 16 vendors 28 chips include NVIDIA, Ascend, ARM, RISC-V AI chipse, Mthread, MetaX etc.

Training developers to master a new programming language
(e.g. Triton-TLE) takes years ...



But now it is AI to write the code, not human.



KernelGen: Agentic system for AI operator auto generation — support various AI chips

— Key for Day0 New model support



KernelGen is an automatic operator generation system for new models on vairoous AI chips, covering the complete lifecycle including **code generation**, **test case generation**, **precision/speed-up testing**, **continue optimization**.

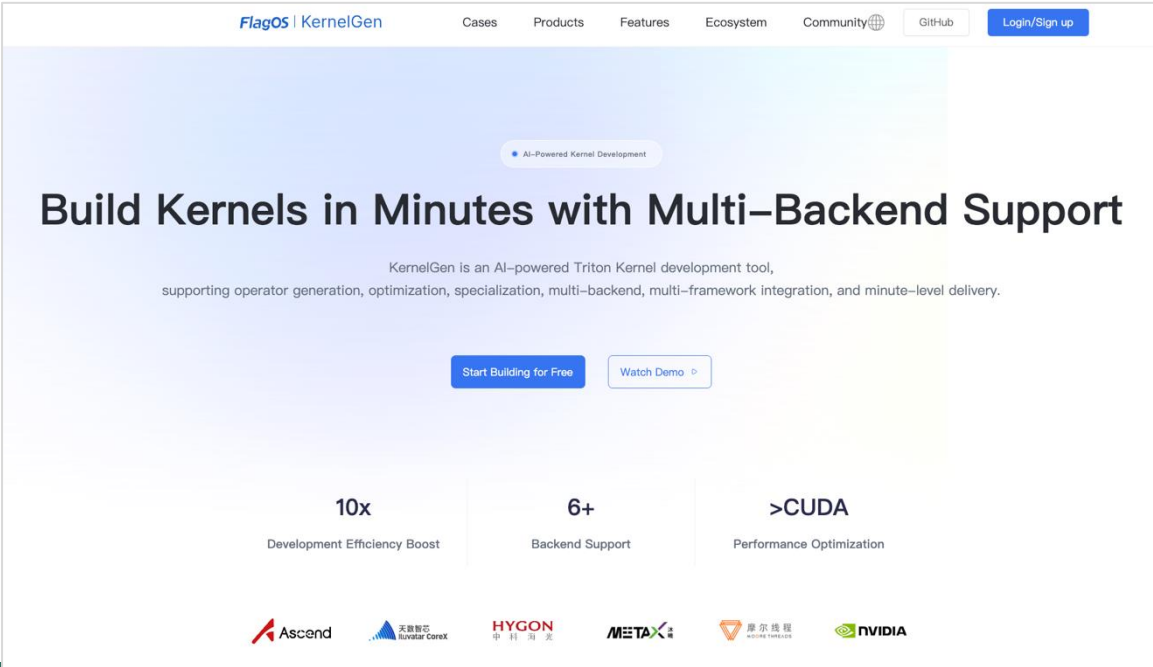
Supported Languages & Chips

Languages: Triton, Triton-TLE

Chips: Huawei Ascend, MThreads, Hygon, iluvatar, MetaX, NVIDIA

Interfaces: VS Code, Claude Code, OpenClaw

Products: Web platform, Agent Skills, MCP



Performance Highlights

- **NVIDIA:** Significantly outperforms Claude Code in correctness and speedup (110 operators on KernelGen-bench)

KernelGen 2.0 vs Claude Code (LayerNorm Example)

Metric	KernelGen 2.0	Claude Code
Execution Time	14m10s ↓	34m42s
Token Consumption	10K ↓	21.6K
Accuracy	100%	100%
Speedup	1.14x ↑	0.64x

- **The other 5 AI chips:** >92% operator generation capability, >50% operators exceed its native performance in C language.

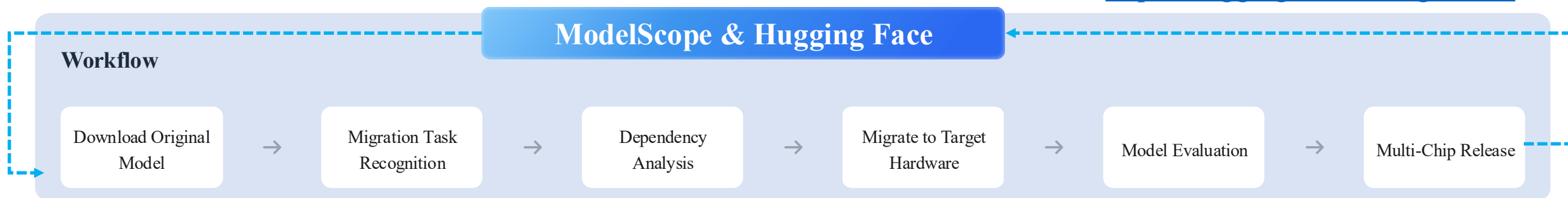
DeepSeek V4 introduced 5 key operators; KernelGen generated 3 Triton kernels:

Kernels	What it does	Gain
FP8 MatMul	128×128 block-wise scaling, fp8e8m0 scale	2.47× DS-V4 original version
Sparse Attention	attention-sink + topk selection	1.89× DS-V4 original version
Hadamard Transform	Butterfly stages, supports non-power-of-two dimensions	1.13× DS-V4 original version

FlagRelease Large Model Cross-Chip Migration & Release Platform —— Release Once, Run Across Chips

One-stop provision of model files, code and images for multiple chips. Developers can launch new AI models on different hardware in three steps, without manual migration or code changes.

<https://huggingface.co/flagrelease>



Supported hardware: Nvidia, ARM, Iluvatar, Metax, Cambricon, Kunlunxin, Hygon, Ascend, PPU, Mthread etc.

The screenshot displays the FlagRelease platform interface, showing a grid of model releases. Each entry includes the model name, size, update time, and a "NEW" badge. The models are organized into two main categories: Qwen3.6-35B-A3B-nomtp and MiniMax-M2.7.

Model Name	Size	Update Time	NEW
FlagRelease/Qwen3.6-35B-A3B-nomtp-iluvata...	36B	Updated about 10 hours ago	
FlagRelease/Qwen3.6-35B-A3B-nomtp-nvidia...	36B	Updated about 10 hours ago	
FlagRelease/Qwen3.6-35B-A3B-nomtp-metax-F...	36B	Updated about 10 hours ago	
FlagRelease/Qwen3.6-35B-A3B-nomtp-zhenwu...	36B	Updated about 10 hours ago	
FlagRelease/Qwen3.6-35B-A3B-nomtp-hygon-F...	36B	Updated about 10 hours ago	
FlagRelease/Qwen3.6-35B-A3B-nomtp-kunlunx...	36B	Updated about 10 hours ago	
FlagRelease/Qwen3.6-35B-A3B-nomtp-ascend...	36B	Updated about 10 hours ago	
FlagRelease/MiniMax-M2.7-iluvatar-Fl...	228.69B	2026.04.15	NEW
FlagRelease/MiniMax-M2.7-nvidia-Flag...	228.69B	2026.04.13	NEW
FlagRelease/MiniMax-M2.7-metax-FlagOS	228.69B	2026.04.12	NEW
FlagRelease/MiniMax-M2.7-hygon-FlagOS	228.69B	2026.04.12	NEW
FlagRelease/MiniMax-M2.7-ascend-Flag...	228.69B	2026.04.12	NEW
FlagRelease/MiniMax-M2.7-zhenwu-Flag...	228.69B	2026.04.12	NEW

FlagOS Day0 Migrated DeepSeek V4 onto different AI Chips

FlagOS enables DeepSeek V4 on 10 different AI chipsets, including hardware without FP4/FP8 and with ≤ 64 GB accelerator memory.

Mode	Chip/Platform	Precision
DeepSeek-V4-Flash	NVIDIA (H20/H100)	BF16 / FP8
	MThread	FP8
	8 AI chips including Ascend, MetaX, PPU/etc.	BF16
DeepSeek-V4-Pro	NVIDIA (H20/H100)	Int8 / FP8
	MThread	FP8
	Hygon/Metax/Ascend	Int8

One-stop code for DeepSeek-V4 with FlagOS

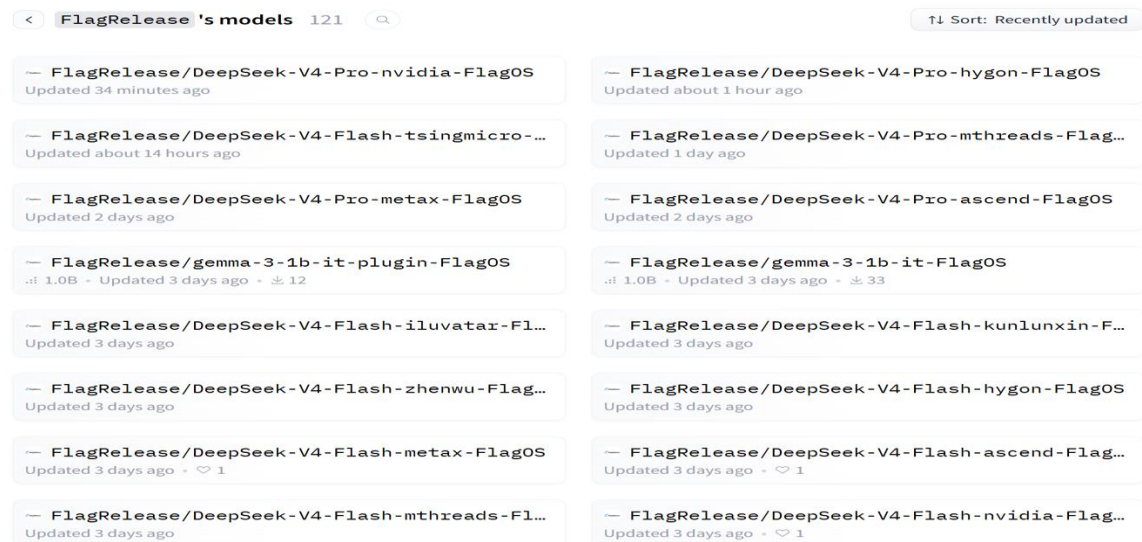
<https://github.com/flagos-ai/DeepSeek-V4-FlagOS>

Inference code and with tools

- O-Groups Grouped Projection Communication
- FP8/FP4 $\rightarrow$ BF16 Weight Conversion Tool
- INT8 MoE Expert Quantization
- Streaming Weight Conversion (Memory-Optimized)

One-stop model files for 10 AI chips:

<https://huggingface.co/flagrelease>



With FlagOS, we have also supported

Qwen3.6 on 7 AI chips · MiniCPM-o 4.5 on 6 AI chips

MiniMax 2.7 on 6 AI chips · DeepSeek V4 on 10 AI chips

in last 3 months

Unified Multi-Chip Plugin System: Inference, Training, and RL Plugins

The unified plugin system turns the “N×M” adaptation problem into “N+M.”

It reduces adaptation cost, enables Day-0 support for new models, and delivers consistent cross-chip results.

Vendor-specific framework patches create inconsistent versions, runtime behavior, parallelism, quantization, and performance.

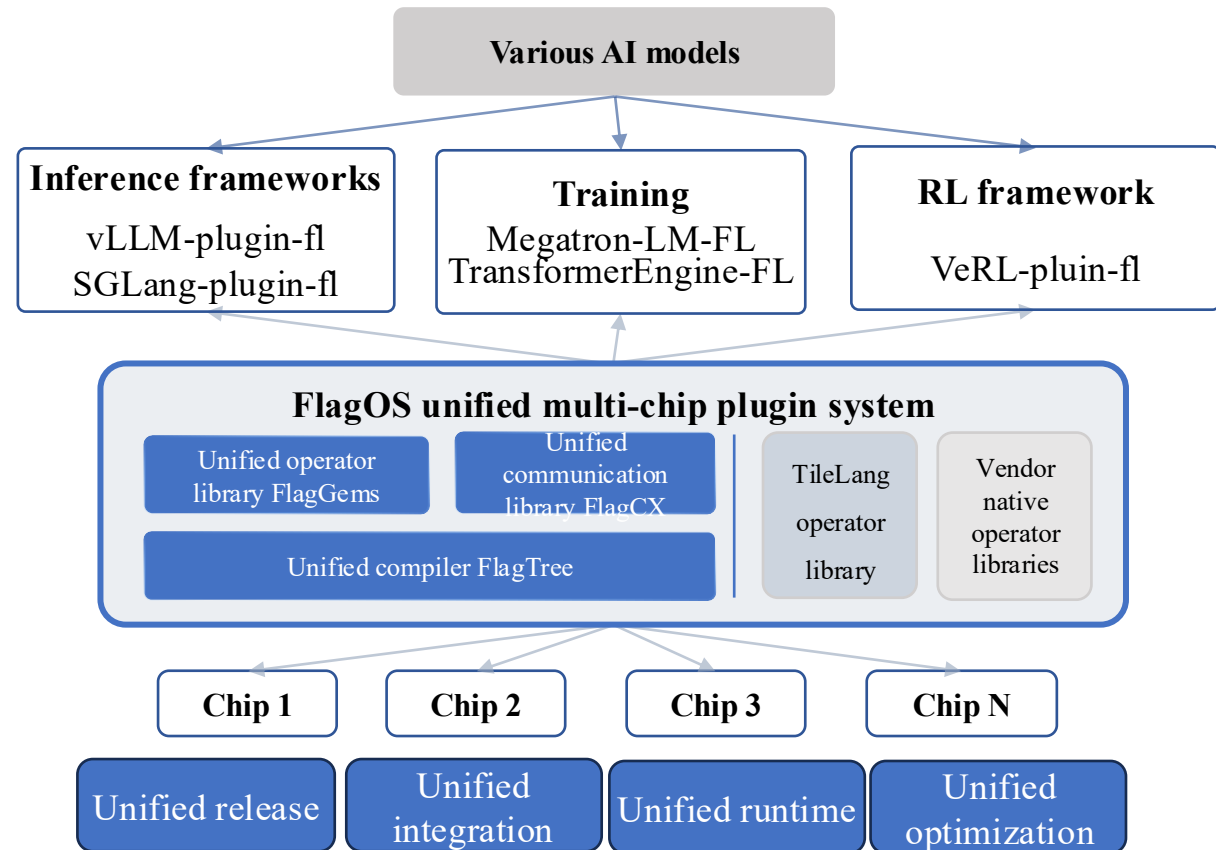
FlagOS plugins provide one unified path for thousands of models across chips.

- Fast adaptation
- Better quality
- Good performance

<https://github.com/flagos-ai/TransformerEngine-FL>

<https://github.com/flagos-ai/vllm-plugin-FL>

<https://github.com/flagos-ai/Megatron-LM-FL>



Upgrading vLLM-plugin-fl to 0.20.0 for DeepSeek v4 multi-chip version enablement. To release SGLang-plugin-fl for next step

Global Open Compute Initiative for AI : We are looking for broader collaboration



In last one year, we bring it together, and make it happening

- UN Office for Digital and Emerging Technologies
- FlagOS community 
- Beijing Academy of Artificial Intelligence 
- Peking University 
- Africa Union – Africa Science and Research Innovation Council
- UNESCO China 



Global Open Compute Initiative for AI: Open Compute turns underutilized AI compute resources worldwide into accessible infrastructure for developing regions, enabling more inclusive global AI innovation.



The first program: China-Africa AI Talent Development Program

BAAI, AU-ASRIC and Peking Univ. together launched the program and invited 11 teachers from different universities from 10 Africa countries. (Apr.15~Apr.30, 2026)



“Open Compute – online lab” powered by FlagOS (>100 AI devices from 3 vendors)



Home Software Stack Tools Docs Learn Event & Competition Online Laboratory

FlagOS Community offers a series of AI courses and welcomes universities/institutions to apply for collaboration with experimental computing support.
Submit Collaboration Application →

The environment is a containerized environment. Once released, all data will be destroyed and cannot be recovered. Please make sure to back up your data in advance.

Welcome, User [User Guide](#)

DevEnvironment Name	Status	Computing Resource Information	Create Time	Release Time	DevEnvironment Access
activity_29jvhy9_8...	Running	GPU:Ascend 910C x 2 Card(s) CPU:20 Core(s) System Storage:200G Memory:80G	2026-04-15 15:55:24 (UTC+8)	2026-04-16 00:00:00 (UTC+8) Countdown: 02 hour 53 minute	Online Laboratory Cloud DevEnvironment Secret Key: zSLJz... Management Enter IDE

An Invitation: Let's Build Open Compute Together

GitHub: <https://github.com/flagos-ai>

<https://flagos.io>